

$$1) D = 4096, F = 4 \cdot D, V = 32,000, L = 64, NH = D$$

Assuming gating einsum type MLP and vanilla MHA ($N=K$)

$$i) \text{ Per layer params} \approx \underbrace{3DF}_{\text{MLP}} + \underbrace{2D(N+K)H}_{\text{attn}} + \underbrace{D}_{\text{layer norm}}$$

$$\begin{aligned} \text{Total} &= L(3DF + 2D(N+K)H + D) + \underbrace{2DV}_{\text{embed+unembed}} \\ &= L(3DF + 4DNH + D) + 2DV \quad (\text{vocab} \cdot \text{hidden dim size}) \\ &= L(3DF + 4D^2 + D) + 2DV \\ &= 64 \cdot (3 \cdot 4096 \cdot 4 \cdot 4096 + 4 \cdot 4096^2 + 4096) + 2 \cdot 4096 \cdot 32000 \\ &\approx 17B \text{ params} \end{aligned}$$

$$ii) \frac{\text{attn params}}{\text{total params}} = \frac{4DNH}{4DNH + 3DF + D} \approx \frac{4D^2}{4D^2 + 12D^2} = \frac{1}{4}$$

$$iii) \underbrace{2}_{\text{keys and vals}} \cdot \underbrace{L}_{\text{\# of layers}} \cdot \underbrace{N}_{\text{kv heads per layer}} \cdot \underbrace{H}_{\text{hidden dim per layer}} = 2LD = 2 \cdot 64 \cdot 4096 \cdot 1 = 524288 \text{ bytes} \approx 524 \text{ kB/token}$$

$$2) A[B_x, D_x] \cdot W[D_x, F]$$

$$\text{mesh} = \{x: 4, y: 8, z: 4\}$$

i) Replicated across Z -axis so each Z -axis "layer" does full matmul.

$$\text{FLOPS}_{\text{Total}} = Z \cdot 2BDF = 2BDFZ$$

ii) Excluding reduction cost from AllReduce,

$$\text{FLOPS}/\text{TPU} = \frac{2BDF}{XY}$$

$$3) A[\textcolor{red}{I}, \textcolor{red}{J}, K, L] \cdot B[\textcolor{red}{I}, \textcolor{red}{J}, M, N, O] \rightarrow C[K, L, M, N, O]$$

$$\text{FLOPS}_{\text{Total}} = 2IJKLMNO$$

4) (Forward pass only, bfl6) $\rightarrow$ factor of $\frac{1}{2}$ from self-attn ignored in big-O

$$\begin{aligned} \text{FLOPS} &= 2BTSKGH + O(BTSN) + 2BTSKGH \\ &= 4BTSNH + O(BTSN) \end{aligned}$$

Comp order: Load Q, K, V from HBM, compute $\text{softmax}(QK^T) \cdot V$ and write back to HBM.

(No loading/storing intermediate results from matrices and softmax).

$$\begin{aligned} \text{Bytes} &= 2 \cdot BTHN + 2 \cdot BSKH + 2 \cdot BSKH + 2 \cdot BTHN \\ &= 4BTHN + 4BSKH = 4BTKGH + 4BSKH \\ &= 4BKH(TG + S) \end{aligned}$$

$$AI = \frac{4BTSNH + O(BTSN)}{4BKH(TG+S)} = \frac{\cancel{4B} + \cancel{S}KH}{\cancel{4B}KH(TG+S)} = \frac{TSG}{TG+S}$$

$$= \frac{TG}{\frac{TG}{S} + 1}$$

During prefill / training time, we compute queries for all tokens ($S=T$):

$$AI = \frac{TG}{\frac{TG}{T} + 1} = \frac{TG}{G+1} \quad (\text{almost always compute bound})$$

During decode (inference-time) we only generate the query for the last token (assuming our kv-cache is full), $T=1$:

$$AI = \frac{G}{\frac{G}{S} + 1} \Rightarrow \lim_{S \rightarrow \infty} AI = G$$

Our # of query groups is small so during decode we're always compute bound.

$$5) \text{ FLOPS}_{\text{self-atten}} = 4BTSNH + O(BTSN)$$

$$\approx 4BTSNH$$

(Assuming vanilla MHA, $K=N$, $NH=D$)

$$\text{FLOPS}_{\text{proj}} = 4BTD(W+K)H$$

$$= 4BTD(2N) \cdot H$$

$$= 8BTD^2$$

$$\text{FLOPS}_{\text{self-attn}} = 4BTS D \geq \text{FLOPS}_{\text{proj}} = 8BT D^2$$

$$\Rightarrow 4BTS D \geq 8BT D^2$$

$$\Rightarrow 4S \geq 8D \Rightarrow S \geq 2D$$

Assuming $D=4096$:

$$S \geq 8192$$

Only at large sequence lengths do attn-flops dominate.

$$6) \text{ Save: } Q[B, T, N, H], K[B, S, K, H], V[B, S, K, H] \\ W_1[B, T, F], W_2[B, T, F], W_3[B, T, D], \\ O[B, T, D]$$

Going backwards through the attn comp graph (figure in section with postnorm):

For $\frac{\partial \mathcal{L}}{\partial W_{\text{out}}}$, we need to rematerialize:

$$O(W_1) * W_2 \Rightarrow O(BTF) \text{ FLOPs}$$

For $\frac{\partial \mathcal{L}}{\partial W_{\text{In1}}}$ and $\frac{\partial \mathcal{L}}{\partial W_{\text{In2}}}$, we need to rematerialize:

$$\text{layernorm}_p (x + O[B, T, D]) \Rightarrow O(BTD) \text{ FLOPS}$$

For $\frac{\partial \mathcal{L}}{\partial W_0}$, need:

$$\text{softmax}(Q \cdot K) \cdot V \Rightarrow 2BTSKGH + O(BTSKG) \\ + 2BTSKGH \text{ FLOPS}$$

$$\approx 4BTSNH \text{ FLOPS}$$

(during training, $T=S$)

$$= 4BT^2NH \text{ FLOPS}$$

For $\frac{\partial \mathcal{L}}{\partial W_Q}$, $\frac{\partial \mathcal{L}}{\partial W_K}$, $\frac{\partial \mathcal{L}}{\partial W_V}$, $\frac{\partial \mathcal{L}}{\partial W_0}$:

No rematerialization needed.

$$\Rightarrow \text{FLOPS}_{\text{Total, remat}} = 4BT^2NH + O(BTD) + O(BTF) \text{ FLOPS}$$

2) Trained for 2.79M H800 hrs on 14.8T tokens.

37B activated params, w/ FP8 FLOPS w/ structured sparsity.

$$\text{FLOPS}_{\text{Total}} = 6.37B \cdot \underbrace{14.8T}$$

repeated tokens (multiple epochs) are included in this count

$$\left(\text{FLOPS/s} \right)_{\text{utilized}} = \frac{6.37B \cdot 14.8T}{2.79M \text{ hrs}} = \frac{(6.37 \cdot 14.8) \cdot 10^{21}}{2.79 \cdot 10^6 \cdot 60 \cdot 60}$$

$$\approx 3.27 \times 10^{14} \text{ FLOPS/s}$$

$$\left(\text{FLOPS/s} \right)_{\text{peak, H800, FP8, no sparsity}} \approx 1.5 \times 10^{15} \text{ FLOPS/s}$$

$$\Rightarrow \frac{3.27 \times 10^{14}}{1.5 \times 10^{15}} = 0.218 \Rightarrow 21.8\% \text{ hardware utilization}$$

8) $X[B, D]$, $E[D, F]$, k activated experts per token

↙ flattened

each expert

int8

$$\text{FLOPS} = 2BDF \cdot k$$

$$\text{bytes} = EDF$$

(need to load all experts and then route each token to selected)

(assuming input is already loaded)

$$T_{\text{comp}} \geq T_{\text{comm}}$$

$$\Rightarrow \frac{2BDF \cdot k}{C} \geq \frac{EDF}{W} \Rightarrow \frac{2BK}{C} \geq \frac{E}{W}$$

$$\Rightarrow B \geq \frac{CE}{2KW}$$

On TPU v5e with $k=8$, $E=256$

$$B \geq \frac{(3.94 \cdot 10^{14}) \cdot 256}{2 \cdot 8 \cdot 8 \cdot (2 \cdot 10^{11})} = 3940$$